Johns Hopkins Engineering

Introduction to GPU Programming

Module 8 Lecture 8A NVidia GPU-Accelerated Libraries – Solve FFTs using cuFFT

Material derived from https://developer.nvidia.com/cufft

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Overview

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2. Performance
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8.1 Intro

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Intro

- “The NVIDIA CUDA Fast Fourier Transform library (cuFFT) provides a simple interface for computing FFTs up to 10x faster. By using hundreds of processor cores inside NVIDIA GPUs, cuFFT delivers the floating-point performance of a GPU without having to develop your own custom GPU FFT implementation.”

- “Widely used in applications ranging from computational physics to image processing and general signal processing, the Fast Fourier Transform is an efficient algorithm for computing discrete Fourier transforms of complex or real-valued data sets. cuFFT uses algorithms based on the well-known Cooley-Tukey and Bluestein algorithms, so you can be confident that you’re getting accurate results faster than ever.”

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Features

- 1D, 2D, 3D transforms of complex and real data types
- 1D transform sizes up to 128 million elements
- Flexible data layouts by allowing arbitrary strides between individual elements and array dimensions
- FFT algorithms based on Cooley-Tukey and Bluestein
- Familiar API similar to FFTW Advanced Interface
- Streamed asynchronous execution
- Single and double precision transforms
- Batch execution for doing multiple transforms
- In-place and out-of-place transforms
- Flexible input & output data layouts, similar to FFTW "Advanced Interface"
- Thread-safe & callable from multiple host threads”

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8.2 Performance

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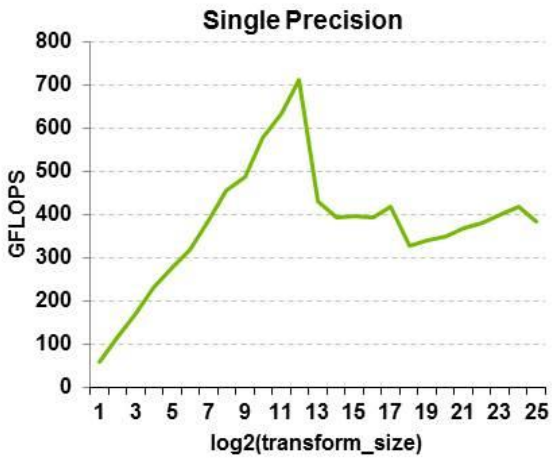
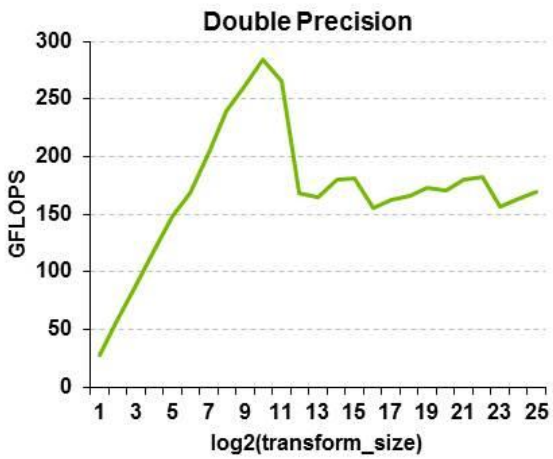
Performance

cuFFT: up to 700 GFLOPS

1D Complex, Batched FFTs

Used in Audio Processing and as a Foundation for 2D and 3D FFTs





Performance may vary based on OS version and motherboard configuration

• cuFFT 6.0 on K40c, ECC ON, 32M elements, input and output data on device

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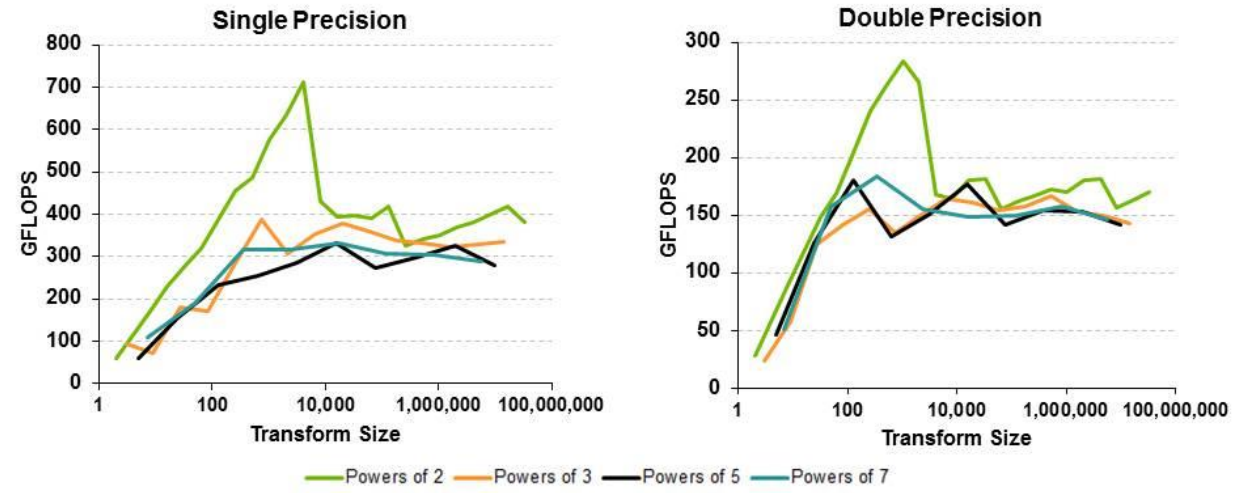
Performance

cuFFT: Consistently High Performance



Performance may vary based on OS version and motherboard configuration • cuFFT 6.0 on K40c, ECC ON, 28M-33M elements, input and output data on device

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8.3 Sample Code

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Sample Code

```
#define NX 64
#define NY 64
#define NZ 128
cufftHandle plan;
cufftComplex *data1, *data2;
cudaMalloc((void**)&data1, sizeof(cufftComplex)*NX*NY*NZ);
cudaMalloc((void**)&data2, sizeof(cufftComplex)*NX*NY*NZ);

/* Create a 3D FFT plan. */
cufftPlan3d(&plan, NX, NY, NZ, CUFFT_C2C);
/* Transform the first signal in place. */
cufftExecC2C(plan, data1, data1, CUFFT_FORWARD);
/* Transform the second signal using the same plan. */
cufftExecC2C(plan, data2, data2, CUFFT_FORWARD);

/* Destroy the cuFFT plan. */
cufftDestroy(plan);
cudaFree(data1); cudaFree(data2);
```

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8.4 Types

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Types

- cuFFT Plan Creation
  - Simple 1, 2, and 3-dimensional FF
    - cufftPlan1D() / cufftPlan2D() / cufftPlan3D()
  - Batched Input and complex data layouts
    - cufftPlanMany()
  - Like above supporting any type of precision
    - cufftXtMakePlanMany()

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Types

- cuFFT Plan Execution
  - Complex-to-Complex Single/Double Precision
    - cufftExecC2C() / cufftExecZ2Z()
  - Real-to-Complex Forward Transformation Single/Double Precision
    - cufftExecR2C() / cufftExecD2Z()
  - Real-to-Complex Inverse Transformation Single/Double Precision
    - cufftExecC2R() / cufftExecZ2D()

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8.5 Data Layout

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Data Layout

FFT type	input data size	output data size
C2C	x cufftComplex	x cufftComplex
C2R	x 2 + 1 cufftComplex	x cufftReal
R2C*	x cufftReal	x 2 + 1 cufftComplex

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Multidimensional Data Layout

Dims	FFT type	Input data size	Output data size
1D	C2C	N 1 cufftComplex	N 1 cufftComplex
1D	C2R	⌊ N 1 2 ⌋ + 1 cufftComplex	N 1 cufftReal
1D	R2C	N 1 cufftReal	⌊ N 1 2 ⌋ + 1 cufftComplex
2D	C2C	N 1 N 2 cufftComplex	N 1 N 2 cufftComplex
2D	C2R	N 1 (⌊ N 2 2 ⌋ + 1) cufftComplex	N 1 cufftReal
2D]	R2C	N 1 N 2 cufftReal	N 1 (⌊ N 2 2 ⌋ + 1) cufftComplex
3D	C2C	N 1 N 2 N 3 cufftComplex	N 1 N 2 N 3 cufftComplex
3D	C2R	N 1 N 2 (⌊ N 3 2 ⌋ + 1) cufftComplex	N 1 N 2 N 3 cufftReal
3D	R2C	N 1 N 2 N 3 cufftReal	N 1 N 2 (⌊ N 3 2 ⌋ + 1) cufftComplex